

Business Requirements Document

US Traffic Risk Intelligence | 2025–2026 Review

Item	Detail
Project	US Traffic Risk Intelligence Dashboard
Prepared by	Aparna Cheekatla
Tools	Power BI Desktop (Power Query, data model, DAX), Power BI Service
Period	1 Jan 2025 – 31 Dec 2026 (Current Year = 2026, Previous Year = 2025)
Data	Simulated US road accident dataset, 307,973 rows × 21 columns
Version	2.0 (final, matches the delivered dashboard)
Date	29 Sep 2026

1. Project Overview

The client needs an interactive Power BI dashboard that compares US road accidents and casualties in 2026 against 2025. It gives road-safety stakeholders one view of accident volume, severity, contributing conditions, high-risk times and locations, so they can target enforcement, road improvements and emergency resources.

2. Business Objectives

1. Report total casualties and total accidents for the current year, with year-over-year (YoY) change.
2. Show casualties by severity (Fatal, Serious, Minor) and how each changed YoY.
3. Show how casualties change month by month compared with the previous year.
4. Identify the road types, speed limits, times of day, areas and light conditions linked to the most (and most severe) casualties.
5. Show where accidents happen on a US map.

3. Stakeholders

#	Stakeholder	Use of the dashboard
1	US Department of Transportation (incl. NHTSA)	Track national safety performance and YoY trends to guide policy and funding
2	State Departments of Transportation	Find road types and speed zones with the most severe casualties to prioritize road improvements
3	Police (state highway patrols, city police departments)	Plan patrols by location, time of day and conditions

#	Stakeholder	Use of the dashboard
4	Emergency Medical Services (EMS)	Plan ambulance coverage by location, severity and peak hours
5	Road safety organizations	Aim awareness campaigns at high-risk speeds, times and conditions
6	Transport operators (bus, trucking, rideshare)	Understand risk by road type and time of day for driver training
7	Traffic management agencies	Improve traffic control on high-risk roads and during rush hours
8	Public	Stay informed about road safety trends
9	Media	Report accurate road safety figures

4. Scope

In scope

- Accident and casualty records for calendar years 2025 and 2026
- KPI cards with YoY, CY vs PY monthly trend, category breakdowns and a US map
- Slicers for road surface and weather conditions
- Data cleaning in Power Query, a data model with a Calendar table, and DAX time intelligence
- Publishing to Power BI Service and building a dashboard

Out of scope

- Real or live accident data (the dataset is simulated)
- Forecasting or predictive models
- Data before 2025 or after 2026

5. Data Source

One Excel sheet, `US_Road_Accident_Data_2025_2026.xlsx` (sheet **Data**), with one row per accident. The data is **simulated**: it follows the structure of the public UK road accident dataset but uses 58 US counties in 31 states, 54 US police agencies, mph speed limits and US road and vehicle terms.

Field	Type	Example values
Accident_Index	Text (unique key)	US00000001
Accident Date	Date	1/1/2025 – 12/31/2026
Day_of_Week	Text	Monday ... Sunday
Junction_Control	Text	Traffic signal, Stop sign, Yield sign or uncontrolled
Junction_Detail	Text	T-intersection, Four-way intersection, Not at intersection
Accident_Severity	Text	Fatal, Serious, Minor

Field	Type	Example values
Latitude / Longitude	Decimal	29.7349 / -95.3913
Light_Conditions	Text	Daylight, Dark - lighted, Dark - not lighted
Local_Authority_(District)	Text	Harris County, TX
Carriageway_Hazards	Text	None, Animal on road, Object on road
Number_of_Casualties	Whole number	1 – 15
Number_of_Vehicles	Whole number	1 – 6
Police_Force	Text	Houston Police Department, California Highway Patrol
Road_Surface_Conditions	Text	Dry, Wet, Snow, Ice, Standing water
Road_Type	Text	Two-lane road, Interstate/Freeway, Divided highway, Roundabout, One-way street, Ramp
Speed_limit	Whole number (mph)	15 – 75
Time	Time	17:25
Urban_or_Rural_Area	Text	Urban, Rural
Weather_Conditions	Text	Clear, Rain, Snow, Fog, Blizzard
Vehicle_Type	Text	Passenger car, SUV, Pickup truck, Semi-truck/Tractor-trailer, Bus

Data quality issues found and handled

- 156 fatal records misspelled as "**Fetal**" in Accident_Severity → corrected to "Fatal"
- **20 blank cells**: Road_Surface_Conditions (8), Road_Type (6), Weather_Conditions (6) → kept in totals and hidden only in the visuals where they would appear as "(Blank)"

6. Business Requirements

ID	Type	Requirement	Delivered as
BR-01	Primary KPI	Total casualties for Current Year and YoY growth	KPI card: 206.1K, -12.0%
BR-02	Primary KPI	Total accidents for Current Year and YoY growth	KPI card: 145K, -11.3%
BR-03	Primary KPI	Casualties by severity for Current Year, each with YoY growth	3 KPI cards: Fatal 2.8K (-23.7%) , Serious 33.8K (-17.3%) , Minor 169.5K (-10.6%)
BR-04	Secondary KPI	Severity mix by speed limit	100% stacked column chart: " Higher Speed, Higher Severity "
BR-05	Trend	Monthly casualties, Current Year vs Previous Year	Area chart: " CY Casualties vs PY Casualties Monthly Trend "

ID	Type	Requirement	Delivered as
BR-06	Breakdown	Current Year casualties by road type	Bar chart: "Casualties by Road Type"
BR-07	Breakdown	Current Year casualties by area (Urban/Rural)	Donut chart
BR-08	Breakdown	Current Year casualties by light condition (Day/Dark)	Donut chart (Light_Conditions grouped into Day / Dark)
BR-09	Location	Casualties and accidents by location	Azure map by county (latitude/longitude)
BR-10	Time of day	Current Year casualties by time of day	Column chart: "Peak Risk Hours" (Night, AM Rush, Midday, PM Rush, Evening)

7. Functional and Non-Functional Requirements

ID	Requirement	Status
FR-01	Road Surface and Weather Conditions slicers filter every visual on the page	Done
FR-02	CY, PY and YoY values are calculated with DAX time intelligence on a Calendar table	Done
FR-03	YoY % shows in green when casualties or accidents fall	Done
FR-04	Tooltips show exact values on hover	Done
FR-05	Clicking a visual cross-filters the others	Done
FR-06	Report is published to Power BI Service and a dashboard is created from it	Done
NFR-01	One consistent color theme, clear titles, readable by non-technical users	Done
NFR-02	The report states that the data is simulated	Done ("Simulated US data · Built in Power BI")

8. Data Preparation (Power Query)

1. Set data types: Accident Date = Date, Time = Time, Latitude/Longitude = Decimal, counts and Speed_limit = Whole number.
2. Replaced **"Fetal"** with **"Fatal"** in Accident_Severity.
3. Kept blank records in all totals; excluded "(Blank)" only in the Road Type chart.
4. Confirmed 307,973 rows and a unique Accident_Index.

Groups created in the model

Group field	Groups
Road_Surface_Conditions (groups)	Dry · Wet (incl. standing water) · Ice/Snow
Weather_Conditions (groups)	Broader weather categories (e.g. Clear, Rain, Snow, Fog, Other)
Light_Conditions (groups)	Day (Daylight) · Dark (all "Dark - ..." values)
Vehicle_Type (groups)	Car, Truck, Bike, Bus, Agricultural, Other vehicle (available for filtering and future analysis)

9. Data Model and DAX

Model

- **Data:** fact table, one row per accident
- **Calendar:** date table (Date, Month, Month Number, Year), marked as date table
- **Relationship:** Calendar[Date] 1 → * Data[Accident Date]

Measures

Measure	DAX
CY Casualties	<code>TOTALYTD(SUM(Data[Number_of_Casualties]), 'Calendar'[Date])</code>
PY Casualties	<code>CALCULATE(SUM(Data[Number_of_Casualties]), SAMEPERIODLASTYEAR('Calendar'[Date]))</code>
YOY casualties	<code>DIVIDE([CY Casualties] - [PY Casualties], [PY Casualties])</code>
CY Accident Count	<code>TOTALYTD(COUNT(Data[Accident_Index]), 'Calendar'[Date])</code>
PY Accident Count	<code>CALCULATE(COUNT(Data[Accident_Index]), SAMEPERIODLASTYEAR('Calendar'[Date]))</code>
YOY Accidents	<code>DIVIDE([CY Accident Count] - [PY Accident Count], [PY Accident Count])</code>

Severity cards reuse CY Casualties and YOY casualties with a visual-level filter on Accident_Severity.

Calculated columns

Table / column	Purpose
<code>Calendar = CALENDAR(MIN(Data[Accident Date]), MAX(Data[Accident Date]))</code>	Date table
Time Band	Groups the hour of each accident into Night (0–6), AM Rush (6–10), Midday (10–15), PM Rush (15–19), Evening (19–24)
Time Band Order	Numeric 1–5, used to sort Time Band

Table / column	Purpose
Weekday No	WEEKDAY(Data[Accident Date], 2) , used to sort Day_of_Week

10. Validation Figures

The dashboard matches these totals from the source file.

Measure	2025 (PY)	2026 (CY)	YoY
Total Accidents	163,249	144,724	-11.3%
Total Casualties	234,170	206,101	-12.0%
Fatal Casualties	3,621	2,763	-23.7%
Serious Casualties	40,869	33,818	-17.3%
Minor Casualties	189,680	169,520	-10.6%

2026 breakdowns

View	Values
Road type	Two-lane road 147K · Interstate/Freeway 17K · Roundabout 14K · Divided highway 14K · One-way street 7K · Ramp 7K
Area	Urban 78.56% · Rural 21.44%
Light	Day 72.65% · Dark 27.35%
Time of day	Night 11.3K · AM Rush 36.4K · Midday 58.4K · PM Rush 65.1K · Evening 34.9K
Fatal share by speed	1.0% below 55 mph · 2.1% at 55 mph and above

11. Acceptance Criteria

- | Loaded row count is 307,973; Accident_Severity holds only Fatal, Serious, Minor
- | BR-01 to BR-10 are on the report and match the validation figures
- | CY, PY and YoY values are correct
- | Both slicers filter all visuals; no "(Blank)" category shows in charts
- | Report is published to Power BI Service and the dashboard opens without errors

12. Change Log

Version	Change
1.0	Initial BRD from client requirements

Version	Change
2.0	Adapted to US simulated data (2025–2026). BR-04 changed from "casualties by vehicle type" to " severity mix by speed limit ", which gives stakeholders a more actionable view. BR-10 (Peak Risk Hours) added. Vehicle type kept as a grouped field in the model.

13. Assumptions

- Current Year is 2026 and Previous Year is 2025.
- The simulated data is treated as complete and accurate for both years.
- Location analysis uses county-level latitude/longitude.